

ALESSANDRA GUERINONI

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EDUCATION

Bachelor of Engineering with Distinction, Mechatronics Engineering

Sept 2016 - Apr 2022

McMaster University

- Awards: Dean's Honour List (multi-time); Golden Key Award (top 15% of the faculty); Future Leader Recognition Award; Faculty of Engineering Entrance Award.

Machine Learning Certificate

May 2024 - Aug 2024

Cornell University

WORK EXPERIENCE

Senior Software Engineer, Tech Lead

July 2022 - Present

Tesla

- Advanced from Associate to Senior Software Engineer in under 3 years (two promotions, Apr 2025).
- Recruited to a critical org-wide engineering platform migration (~\$100M in projected savings); shipped P1 bug fixes that prevented company-wide design work from halting, coordinated cross-team infrastructure, and unblocked change management for design data.
- Helped build a Kafka/Kubernetes/Go notification pipeline that gave multiple teams real-time access to engineering data from the platform, replacing manual polling and unlocking downstream automation; \$18.8M in validated business impact.
- Led a team to ship a full-stack "GitHub for engineering design data": a change review and release platform with 3D geometry and 2D PDF diff (Next.js, React, Go microservices, Kafka, S3); delivered 10X performance, 99.76% uptime, and a ~20-to-8 day approval cycle (60% reduction); \$3.1M/year validated savings.
- Spearheaded AI integration into the engineering change review workflow, shipping models that classify design changes at 97% accuracy and eliminate ~6,000 hours/year of manual review of design changes (\$1.2M/year saved).
- Helped build ETL pipelines (Go, MySQL, Temporal) exposing engineering data to internal services via gRPC and REST; ~\$10M in validated business impact.
- Own production reliability for team platform services (on-call, observability with Prometheus/Grafana/Splunk); routinely pulled in after hours to debug cross-team production incidents.
- Defined authorization architecture (OAuth, M2M, on-behalf-of flows) for platform services and negotiated API contracts with consumer teams.
- Mentored engineers, drove hiring through technical screens and panel interviews, and sponsored an intern through to full-time conversion.
- Acting Product Manager since Apr 2026, owning backlog prioritization, product strategy, and stakeholder alignment after PM departure.

Project Manager, Governance & University Collaboration

May 2019 - Aug 2020

Bombardier, Mitsubishi Heavy Industries

- Managed 50+ enterprise projects; led a team of 6 coordinating university partnerships and recruiting 26 interns; negotiated contracts reducing cost overhead by 2%.
- Built Python tooling for Oracle database access to streamline project reporting.

Teaching Assistant, Discrete Mathematics & Computational Thinking

Jan 2019 - Apr 2019 & Sept 2021 - Dec 2021

McMaster Faculty of Engineering

- Facilitated Haskell and computer science learning for 50+ students; led tutorials, graded assignments, and ran consultations.

Software Engineering Camp Counselor

Apr 2018 - Sept 2018

McMaster Faculty of Engineering

- Introduced 250+ children to Software Engineering through Python, Arduino, and web projects; delivered STEM lectures to 100+ students across Hamilton.

PROJECTS

TriAuto: Automated Emergency Triage System

Sept 2021 - Apr 2022

McMaster University Capstone

- Co-developed with a 4-person team a real-time clinical decision-support system for emergency department triage that ingests patient intake data and produces a recommended acuity classification (CTAS 1 to 5), giving nurses a second-check during overwhelmed shifts when mis-triage risks delaying care. Stack: React, Node.js, MySQL, Socket.IO.

SKILLS

Languages:

Go • TypeScript • JavaScript • Python • Java • C++ • SQL

Backend & Data:

Kafka • Temporal • MySQL • Redis • gRPC • Protobuf • REST • OAuth • JWT

Cloud & DevOps:

AWS • Azure • Docker • Kubernetes • S3 • Jenkins • ArgoCD • GitHub Actions • Prometheus • Grafana • Splunk

Frontend:

Next.js • React • Node.js